



Predictors of Adolescent Sleep Duration

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RESEARCH QUESTION

? How do marijuana use, fruit consumption, and television watching affect adolescent sleep duration?

VARIABLES

Independent Variables (IVs)

 Marijuana Initiation (Early Use vs. Never)	Binary
 Fruit Eating (Regular vs. No)	Binary
 Television Watching (Daily vs. No)	Binary

Dependent Variable (DV)

 Sleep Duration (Hours per Night, 4–10)	Continuous
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METHOD

Multiple Linear Regression (OLS)

$$\text{Sleep} = \beta_0 + \beta_1(\text{Marijuana}) + \beta_2(\text{Fruit}) + \beta_3(\text{TV}) + \epsilon$$



Goal:
Estimate the independent impact of each behavior on sleep duration.

KEY RESULTS (EFFECT ON SLEEP)



Marijuana Initiation



0.37 hrs
(22 min)



Fruit Eating



0.22 hrs
(13 min)



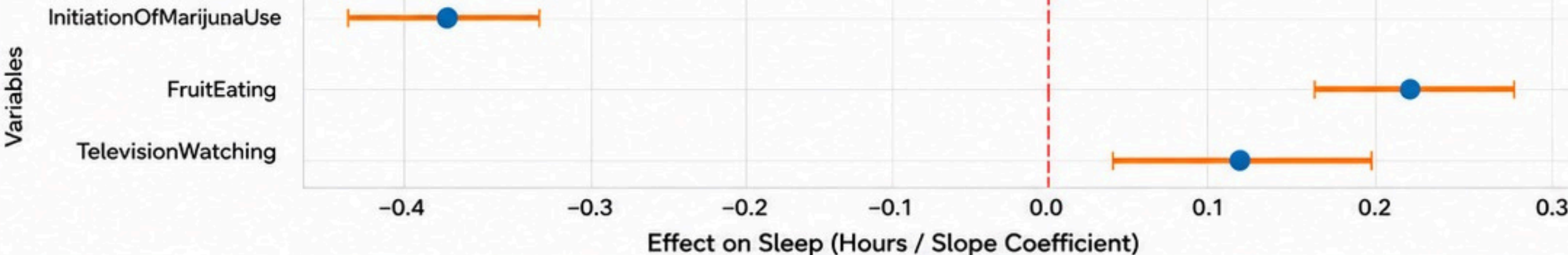
Television Watching



0.11 hrs
(7 min)

★ All predictors are statistically significant ($p < 0.05$)

PREDICTORS OF ADOLESCENT SLEEP DURATION (95% CONFIDENCE INTERVALS)



INTERPRETATION



Early marijuana use is associated with **22 fewer minutes of sleep per night**, the strongest effect.

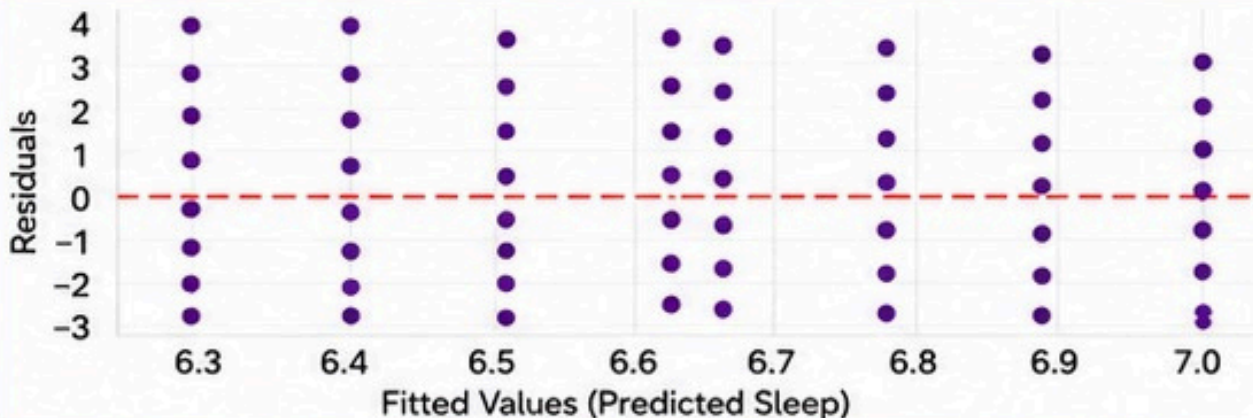


Regular fruit consumption is associated with **13 more minutes** of sleep per night.

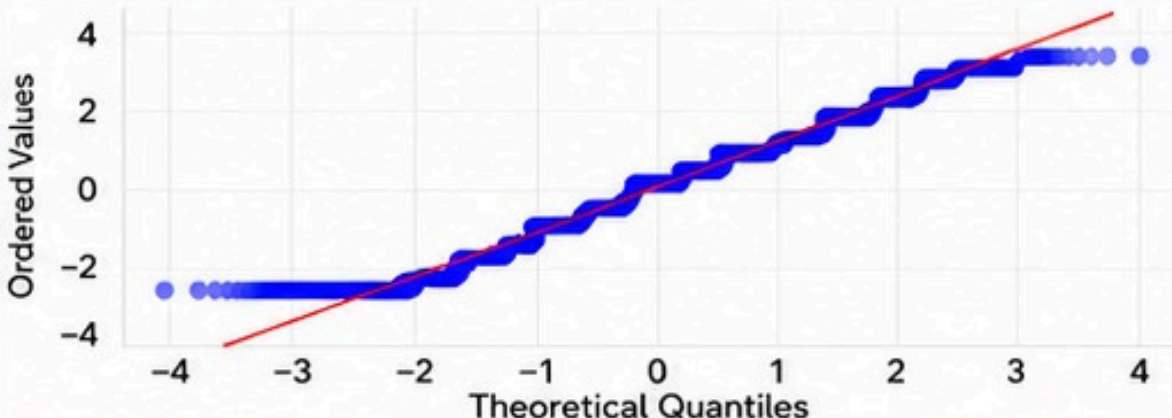


Daily TV watching is associated with **7 more minutes** of sleep per night.

DIAGNOSTIC CHECK: RESIDUALS VS FITTED VALUES



DIAGNOSTIC CHECK: NORMAL Q-Q PLOT



CONCLUSION



- All three behaviors significantly predict adolescent sleep duration.
- Early marijuana use has the largest negative effect (22 fewer minutes).
- Healthy behavior (fruit eating) is linked to longer sleep.
- The model is statistically significant but explains only a small portion of sleep variation.



DATA SOURCE
YRBS 2007



SAMPLE SIZE
n = 11,946

R²
0.023

F-STATISTIC
93.28

p-VALUE
< 0.001



The model is statistically significant but explains only 2.3% of the variation in sleep duration.